DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
(An Institution of National Importance under Ministry of Education, Govt. of India)
MID SEMESTER EXAMINATIONS-II
B. Tech I Semester
MAIC-101: Differential Calculus and Differential Equations

**Date:** 23-10-2024 **Time:** 08:30 AM to 09:20AM **Max. Marks:** 20

**N.B.:** Answer ALL questions. Each question carries equal marks.
Answers to all parts of each question should be in one place.

| | |
|---|---|
| **Q1** | If $f(x,y) = \begin{cases} \frac{x^3-y^3}{x^2+y^2} : (x,y) \neq (0,0) \\ 5 \quad : (x,y) = (0,0) \end{cases}$. Find out the type of discontinuity of $f$ with explanation. |
| **Q2** | The temperature $T$ at any point $(x,y,z)$ in space is $T(x,y,z) = kxyz^2$ where $k$ is a non-zero constant. Find the highest temperature on the surface of the sphere $x^2+y^2+z^2 = a^2$. |
| **Q3** | If $u(x,y) = x\, f_1\left(\frac{y}{x}\right) + f_2\left(\frac{y}{x}\right)$ then find out $xu_x + yu_y$. |
| **Q4** | Solve $\frac{d^3x}{dy^3} - 3\frac{d^2x}{dy^2} - \frac{dx}{dy} + 3x = 0$. |

********